Status	Finished
Started	Thursday, 10 September 2026, 1:55 PM
Completed	Thursday, 10 September 2026, 2:06 PM
Duration	10 mins 55 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Complete

Mark 1.00 out of 1.00

Which statement is the objective of bandwidth utilization?

- a. Bandwidth utilization is about effective use of time division multiplexing
- b. Bandwidth utilization is about effective use of frequency division multiplexing
- c. Bandwidth utilization is about the calculation of bit/signal ratio.
- d. Bandwidth utilization is about the calculation of bit rate of a channel.
- e. Bandwidth utilization is about the effective use of available bandwidth.

The correct answer is: Bandwidth utilization is about the effective use of available bandwidth.

Question 2

Complete

Mark 1.00 out of 1.00

Fill in the blanks.

In a multiplexed system, _____ signals share the bandwidth of _____data link.

- a. 1; 1
- b. 1; n
- c. n; n
- d. n: 1

The correct answer is: n: 1

Question 3

Complete

Mark 1.00 out of 1.00

Which type of signal transmission is supported by each channel of a link running Frequency Division Multiplexing?

a. Manual Transmission
b. Digital Transmission
c. Automatic Transmission
d. Space Transmission
e. Analog Transmission

The correct answer is:
Analog Transmission

Question 4

Complete

Mark 1.00 out of 1.00

Five channels, each with a 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link in KHz if there is a need for a guard band of 5 KHz between the channels to prevent interference?

Answer only the number

Answer: 520

5 * 100 + (5-1)*5 = 500+20=520 KHz

The correct answer is: 520

Question 5

Complete

Mark 1.00 out of 1.00

Given that a computer is sending data at the rate 1.25 Mbps, and the baud rate 250 Kbaud/s, which QAM does it use?

Answer: 32 -QAM.

1.25 Mbps = 1250 Kbps

1250 Kbps /250 Kbaud/sec = 5 --> 5 bits/baud -->2^5 == 32-QAM

Question 6

Complete

Mark 3.00 out of 3.00

Five data channels (digital), each transmitting at 8 Mbps, use a satellite channel of 10 MHz. Design an appropriate configuration using FDM.

Fill in the blanks. (only number)

- The satellite channel is analog. We divide it into 5 channels, each channel having a 2 MHz bandwidth.
- Each digital channel of 8 Mbps is modulated such that each 4 bits are modulated to 1 Hz.
- One solution is
  - 4-QAM
  - 8-QAM
  - 16-QAM
  - 32-QAM

The correct answer is: 16-QAM

Hint: Slide 14

- The satellite channel is analog. We divide it into 5 channels, each channel having a 10/5=2 MHz bandwidth.
- Each digital channel of 8 Mbps is modulated such that each (8Mbps/2MHz) == 4 bits are modulated to 1 Hz.
- One solution is 16-QAM

Question 7

Complete

Mark 1.00 out of 1.00

Our FDM system has 3 digital data channels, each transmitting at 8 Mbps, 16-QAM is used for modulation. How much total bandwidth is required in MHz?

Hint: Slide 16

Answer only the numbers

Answer: 6

Total data rate = 3*8 Mbps = 24 Mbps

16-QAM = 4 bits / baud

Baud rate = 24 Mbps / 4 bits / baud = 6 Mbauds/s ==> 6 MHz

The correct answer is: 6

Question 8

Complete

Mark 1.00 out of 1.00

CWDM can support longer transmission distance than DWDM.

Hint: Slide 32

- True
- False

The correct answer is 'False'.